- Peritrichous: Flagella all around. Eg. *E.coli.*
- Atrichous: Without any flagella.
  Eg. *Corynebacterium diptherae.*

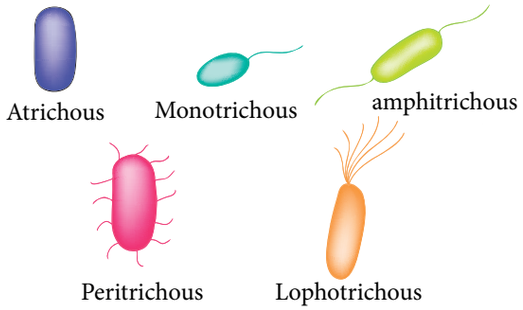



**Figure 16.4** Flagellation types in Bacteria

Bacteria get their food in many ways. Photosynthetic bacteria make their own food (Eg. Cyanobacteria). Bacteria that live in harsh environment use chemicals (Ammonia, Hydrogen sulphide) to produce their food instead of utilizing energy from the sun. This process is called chemosynthesis. Some bacteria exhibit symbiotic relationship (Eg. E.coli lives in the intestine of man). Bacteria reproduces by fission (Binary and multiple fission).

### Activity 1

Take one or two drops of butter milk on a slide and spread it. Heat the slide slightly on a lamp ( 3 – 4 seconds). Add a few drops of crystal violet and leave it for 30 to 60 seconds. Then wash the slide gently with water. Observe the slide under the compound microscope.

## 16.3 Fungi

Fungi are a group of eukaryotic organisms that lack chlorophyll. They grow in dark environments. They may be either unicellular (Eg. Yeast) or multicellular (Eg. Penicillium). They are found in all kinds of habitats. They are included under kingdom Fungi. The study of fungi is called **mycology**. Some fungi are macroscopic (Eg. Mushroom). There are around 70,000 species of fungi, living in the world.

Here, we will study about yeast which is a unicellular fungi. You will study about multicellular fungi elaborately in Chapter 17.

### 16.3.1 Cell structure of Yeast

Yeasts are found freely in the atmosphere. Yeast grows in all kinds of media containing sugar. The cell is ovoid in shape, containing cell wall and a nucleus. The cytoplasm is granular, and has vacuoles, organelles and glycogen an oil globules. Yeast aids in fermentation with the help of the enzyme zymase. It respires anaerobically and reproduces by budding.

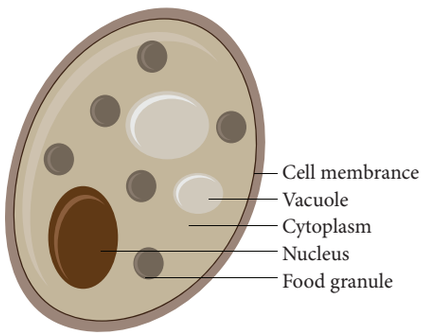



**Figure 16.5** Structure of Yeast

## 16.4 Algae

Algae are very simple plant like eukaryotic organisms. They are found in moist habitats. They are rich in chlorophyll and can be seen as thin film on the surface of lakes and ponds. Therefore they are known as 'grass of water'. They are autotrophic and they produce their own food with the help of chloroplast. Chloroplast contain chlorophyll (green pigments) for photosynthesis. The study of algae is called **algology (phycology)**.

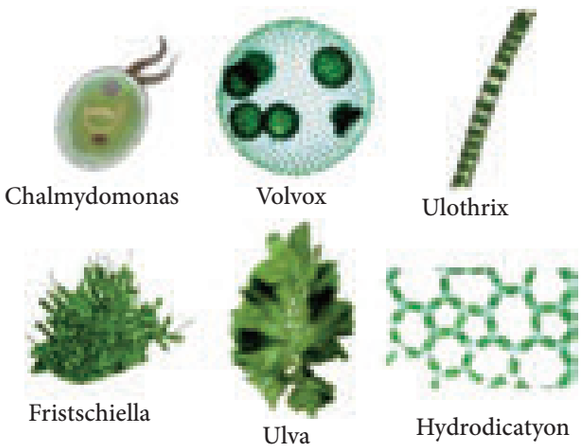



**Figure 16.6** Different types of Algae



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